

nate ripples and get accurate -9V DC. Capacitors C7 and C8 provide additional noise-filtering. The output from voltage regulator IC2 is -9V.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the 9V battery inside the cabinet. At the front panel, connect

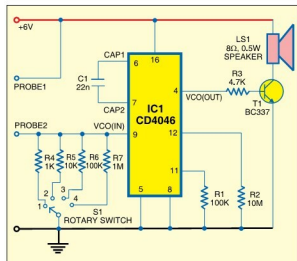
three terminals for outputs.

EFY note. This power supply can be used only for low-power applications. ●

AUDIBLE RESISTANCE-CHECKER

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When measuring the resistance between two points in a circuit, you have to watch the resistance meter while ensuring that the probes are in good contact with both the test points.



This circuit allows you to concentrate on one action, i.e., placing the probes at the right points, while providing an audible assessment of the resistance.

The resistance-checker provides an easy way to determine one of the following three resistance levels between its test points:

Short-circuit. The two points are connected, or there is a very low resistance between them.

Resistance. There is resistance between the two points.

Open circuit. There is no electrical connection between the two points.

The first and the third levels are the

ones that we normally look for. When wiring a circuit, you need to ensure that all the components that should be connected to the power rail are actually connected to it. For example, if a circuit has several logic integrated circuits (ICs), it is to be ensured that all the V_{CC} pins are connected to the positive rail.

Using this checker, one probe can be placed on the positive supply input terminal. You can then run through all the ICs, touching V_{CC} pins with the other probe. If these are connected to the power rail, a high-pitched tone is heard every time. The connections to the negative or ground rail can be checked in the same manner.

The continuity test can also be applied to check undesirable short-circuits. A short-circuit between the positive and ground lines may lead to over-heating, burning smell, discharging of batteries, and too-low (or zero) voltage on the positive rail. To find this kind of short-circuit, first disconnect the circuit from the power supply. Connect the probes to positive and ground terminals. A high-pitched tone of around 925 Hz will be heard. Similarly, around 15Hz buzzing sound is heard when checking for open-circuits (open wires, diodes or transistors with open junction or dry solder).

At the heart of the circuit is IC 4046 phase-locked loop (PLL), which consists of a voltage-controlled

oscillator (VCO), a source follower and two phase comparators. Only VCO pin has been used in this circuit. The frequency of the squarewave output at pin 4 of IC1 depends on the voltage applied to its control input pin 9 and the values of capacitor C1 and resistor R1. If the voltage is equal to half of the supply voltage (+3V in this case), the output frequency is given by:

$$f = 1 / (R1.C1)$$

With the values given here, $f = 455$ Hz.

If the input is reduced to 0V, the frequency too falls to zero. If the input is increased to the supply voltage, the frequency will be 2f, which in this case is 910 Hz.

To avoid complete silence when the input is 0V, resistor R2 offsets the frequency range. A high value of R2 means that there is only a small offset. So 0V produces a low-pitched 15Hz buzz, which is just enough to confirm that the circuit is actually operating. At the other end of the range, 6V produces a note of 910 Hz.

The resistance being checked is connected between the positive rail and one of the many resistors. Range switch S1 lets you select which of the resistors will be in the circuit. On any range, an open circuit between the probes allows the switched-resistor to pull input pin 9 of IC1 down to 0V resulting in a low-pitched buzz.

Also on any range, a short-circuit between the probes pulls the input pin 9 to 6V and a high-pitched tone is heard. To understand what happens when an intermediate resistance is

